Sharon Musa

MSC COMPUTER SCIENCE · YORK UNIVERSITY

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Education

Master of Science (MSc), Computer Science (Thesis) *Sept 2024 – Aug 2026(Expected)*
LASSONDE SCHOOL OF ENGINEERING – YORK UNIVERSITY *Toronto, ON*

Supervisor: Prof. Konstantinos G. Derpanis.
Courses: ML Theory, Probabilistic Models and ML, Data Mining, Deep Learning in CV, Generative Modeling.

Honours Bachelor of Science, Computer Science *June 2024*
LASSONDE SCHOOL OF ENGINEERING – YORK UNIVERSITY *Toronto, ON*

Graduated First Class Honours, with Distinction
Honours Thesis Project: Audio-Video Scene Recognition, **Supervisor:** Prof. Richard P. Wildes

Publications

Sharon S. Musa, Fereshteh Forghani, Harrish Thasarathan, Sonia Joseph, Matthew Kowal, Konstantinos G. Derpanis. "What, Where, and How: Probing Spatiotemporal Representations in Video Foundation Models." Mechanistic Interpretability Workshop, ICML 2026 (Virtual Poster).

Experience

AI Data Trainer/Evaluator – General Stream *Oct. 2025 – Present*
COHERE *Toronto, ON*

- Labeled, ranked, and audited multimodal data for training LLMs and other large-scale models.
- Evaluated outputs across formats (JSON, CSV/TSV, Markdown) while adhering to annotation and quality standards.
- Provided structured feedback to improve dataset quality, annotation guidelines, and downstream model performance.

Graduate Research Assistant – CVIL Lab *Sept. 2024 – Present*
YORK UNIVERSITY – LASSONDE SCHOOL OF ENGINEERING *Toronto, ON*

- Conducted interpretability research on large-scale vision foundation models (e.g., V-JEPA, VideoMAE), analyzing how spatio-temporal properties emerge across network depth.
- Built PyTorch-based pipelines for feature extraction, linear probing, hyperparameter search, and representation analysis across large-scale video benchmarks.
- Trained and evaluate models using multi-GPU distributed PyTorch workflows on Slurm-managed HPC clusters, working with CUDA-enabled GPU infrastructure.
- Analyzed latent geometry in layer representations, including layer-wise probing and geometry-aware steering experiments.
- First-authored workshop paper accepted as a virtual poster at the Mechanistic Interpretability Workshop @ ICML 2026.

Research Intern – IVRL Lab *June. 2025 – Aug 2025*
ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE (EPFL) *Lausanne, Switzerland*

- Summer@EPFL program, supervised by Prof. Raphaël Achddou and Prof. Quentin Bammey under Prof. Sabine Süsstrunk.
- Conducted research on annotated 3D scene reconstruction from video using state-of-the-art vision models including DUSt3R, Fast3R, and VGGT, integrating pose estimation, multi-view alignment, and reconstruction.
- Experimented with Gaussian Splatting–based reconstruction for high-fidelity scene modeling.

Machine Learning Fellow – AI4Good Lab *May. 2024 – June 2024*
MILA, VECTOR INSTITUTE, AMII *Toronto, ON*

- Engaged with curriculum material, gaining expertise in machine learning fundamentals, including neural networks, NLP, and reinforcement learning through 25+ hours of expert-led lectures.
- Developed ML prototypes for social good in collaborative teams, using Python, PyTorch, and other ML frameworks, with 30+ hours of project mentorship and guidance from academic and industry professionals.
- Gained hands-on experience in data preprocessing, model selection, and optimal hyperparameter tuning.
- Built OvaTech AI, an application aimed at improving support for women with polycystic ovary syndrome (PCOS).

Research Assistant – Vision Lab *May. 2023 – May 2024*
YORK UNIVERSITY – LASSONDE SCHOOL OF ENGINEERING *Toronto, ON*

- Engaged in a project focused on deep learning and computer vision techniques for audio-video scene recognition.
- Reviewed literature (i.e., CNNs, Transformers etc.) research papers, and augmented the lab's large-scale datasets.
- Designed and trained deep learning models using Python, TensorFlow, OpenCV, Pandas, etc. to improve recognition accuracy and efficiency.
- Presented project at the Lassonde Undergraduate Research Conference. *Awarded 2nd Best Oral Presentation.*

XR Developer & Research Assistant – Immersive Storytelling Lab *May. 2022 – August 2024*
YORK UNIVERSITY – SCHOOL OF ARTS, MEDIA, PERFORMANCE AND DESIGN *Toronto, ON*

- Developed workflow for moving captured volumetric data postproduction in VR and AR using Depthkit software.
- Collaborated with a diverse team of artists and engineers to highlight social issues and advance digital literature.
- Developed VR and AR projects, resulting in successful deployment at exhibitions and conferences.

Selected Projects

Concept Discovery and Naming with Sparse Autoencoders (SAEs) – (EECS 6322: Deep Learning in CV)

- Implemented mechanistic interpretability and reproduced results from the paper Discover-then-Name for automated concept discovery in vision models.
- Built a PyTorch pipeline to train SAEs on the CC3M dataset and automatically assign semantic labels to discovered concepts by matching decoder dictionary vectors with CLIP embeddings.
- Visualized learned concepts using top-activating images across datasets (ImageNet, CIFAR-10, CIFAR-100), revealing interpretable features from low-level patterns to semantic objects.

Vision Model Interpretability with Sparse Autoencoders – (EECS 6327: Probabilistic Models and ML)

- Investigated internal representations of vision models by training Top-K SAEs on MobileViT activations.
- Learned sparse representations achieving ~0.8 reconstruction accuracy with ~95% sparsity.
- Visualized concept attribution maps showing hierarchical visual features from low-level edges to semantic patterns.

Bias and Fairness in Educational Prediction Models – (EECS 6142: Data Mining)

- Analyzed bias in prediction models using an educational dataset to evaluate fairness across demographic groups.
- Built a Random Forest pipeline with bias mitigation techniques, including SMOTE (pre-processing), adversarial debiasing during training, and post-processing threshold optimization using Fairlearn.
- Evaluated model fairness using metrics such as demographic parity, disparate impact, and equal opportunity, demonstrating that combining mitigation methods improves fairness while maintaining predictive performance.

Skills

Technical:	Python, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, Matplotlib, Numpy, HuggingFace, Linux, Java, HTML/CSS, A-frame, VR/AR, MS Office, Linux, CUDA
Soft:	Leadership, Communication, Teamwork/Collaboration, Fast learning, Research, Creativity
Research Areas:	Mechanistic Interpretability, AI Safety, Computer Vision, Video Foundation Models, AI Safety

Awards and Achievements

- Connected Minds Masters Scholarship ($10k/yr for 2 years)
- Member of the Lassonde Circle of Scholars
- Lassonde Undergraduate Research Award ($10,000)
- Lassonde Scholarship ($15,000/yr for 4 years)
- EPFL Research Program Scholarship (CHF1,800/month for 3 months)
- Black in AI CVPR Student Financial Aid (2023)
- CIFAR Inclusive AI Scholarship ($3,750).

Professional Development

- BlueDot Impact, Technical AI Safety Course *June 2026*
- Mila - Quebec AI Institute, AI4Good Lab 2024 *July 2024*
- Vector Institute, Excel to AI Certificate *Sept 2023*
- Coursera, Deeplearning.AI Introduction to TensorFlow for AI, ML, and Deep Learning *June 2023*
- IEEE/CVF Computer Vision and Pattern Recognition Conference, 2023, Student Attendee *June 2023*